SO(2)-Equivariant Reinforcement Learning for Closed-Loop Manipulation

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Motivation

Robot manipulation from overhead images has a planar rotation symmetry: rotating the workspace yields an equivalent task with a rotated optimal action. **Equivariant SAC** encodes this symmetry in the network weights instead of augmenting the training distribution.

$$\begin{aligned} \text{policy equivariance} \quad & \pi(g{\cdot}s) = g{\cdot}\pi(s) \\ \text{critic invariance} \quad & Q(g{\cdot}s, g{\cdot}a) = Q(s,a) \\ \text{reward invariance} \quad & R(g{\cdot}s, g{\cdot}a) = R(s,a) \\ \text{transition invariance} \quad & P(g{\cdot}s' \mid g{\cdot}s, g{\cdot}a) = P(s' \mid s,a) \end{aligned}$$

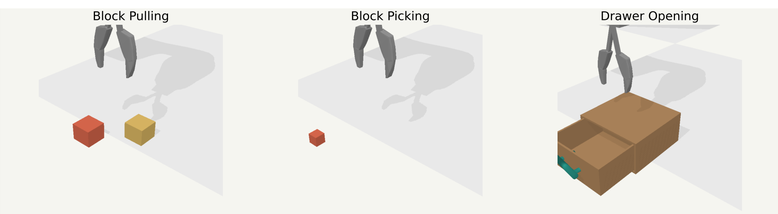
with $g \in C_8$, the cyclic group of eight planar rotations (a discretization of $SO(2)$).

Problem: group-invariant MDP

- $\mathcal{S}$ — $128{\times}128$ depth heightmap concatenated with a tiled scalar gripper-state channel, giving an observation of shape $(2, 128, 128)$.
- $\mathcal{A}$ — continuous 5-DoF in $[-1,1]^5$, decoded to $(\Delta x, \Delta y, \Delta z, \Delta\theta, a_{\text{grip}})$ with $\Delta_{\text{pos}} = 5\,\text{cm}$ and $\Delta_{\text{rot}} = \pi/8$.
- $\mathcal{P}, \mathcal{R}$ — PyBullet rigid-body dynamics; sparse binary reward, $r_t{=}1$ on task success and 0 otherwise.
- γ, H — discount 0.99, horizon 50 env steps.

Reward invariance. R is g-invariant: $R(g{\cdot}s, g{\cdot}a) = R(s,a)$ for all $g \in C_8$.
Transition invariance. P is g-equivariant: $P(g{\cdot}s' \mid g{\cdot}s, g{\cdot}a) = P(s' \mid s,a)$ for all $g \in C_8$.

Experiments



(a) Block Pulling (b) Block Picking (c) Drawer Opening

Three closed-loop manipulation tasks from the paper: drag a block across the table, pick a block off the table, and open a drawer to a target angle. Same observation, action space, and reward structure in all three.

Equivariant SAC

1. Initialize C_8-equivariant actor π_θ and twin critics Q_{ϕ_1}, Q_{ϕ_2} in e2cnn.
2. Target nets $\bar{\phi}_i \leftarrow \phi_i$; warm-start replay $\mathcal{D}$ with scripted expert transitions.

for each env step $t = 1, \ldots, T$:

3. Sample $a_t \sim \pi_\theta(\cdot \mid s_t)$, step env, push $(s_t, a_t, r_t, s_{t+1}) \to \mathcal{D}$.
4. Minibatch $\mathcal{B} \sim \mathcal{D}$; TD target $y = r + \gamma\Big(\min_i Q_{\bar{\phi}_i}(s', a') - \alpha \log \pi_\theta(a' \mid s')\Big)$.
5. Critic update: $\phi_i \leftarrow \phi_i - \eta\, \nabla_{\phi_i}\big(Q_{\phi_i}(s,a) - y\big)^2$.
6. Actor update: $\theta \leftarrow \theta + \eta\, \nabla_\theta\big(\min_i Q_{\phi_i}(s,a) - \alpha \log \pi_\theta(a \mid s)\big)$.
7. Temperature: tune α toward target entropy $\bar{\mathcal{H}} = -|\mathcal{A}|$.
8. Polyak: $\bar{\phi}_i \leftarrow \tau\phi_i + (1-\tau)\bar{\phi}_i$. Only **line 1** changes between Equi SAC and vanilla CNN SAC.

Every loss, target, and update rule is standard SAC (Haarnoja et al., 2018); the symmetry lives in the layers.

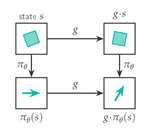
Architecture and hyperparameters

Encoder. Seven-block C_8-equivariant CNN in e2cnn (`R2Conv`, $gspace{=}\text{Rot2dOnR2}(N{=}8)$). Input uses the *trivial representation* (depth and gripper channels are scalars under rotation); hidden layers use the *regular representation*.
Actor head. Final $1{\times}1$ `R2Conv` outputs one **irrep(1)** field for $(\Delta x, \Delta y)$ and eight trivial fields for the three invariant action means $(a_{\text{grip}}, \Delta z, \Delta\theta)$ plus five log-std parameters.
Critic head. Twin heads each wrap the action as irrep$(1)\oplus$trivial^3, mix it into the state feature map with a $1{\times}1$ `R2Conv`, and pool to a single C_8-invariant scalar.

group	C_8	γ	0.99
hidden width	128	τ	0.005
batch size	64	learning rate	10^{-3}
replay buffer	100k	α (init)	0.1, learned
warmup transitions	1k expert	target entropy	-5
env steps / run	20k	eval every	500 steps
eval episodes	5	seeds per cell	2

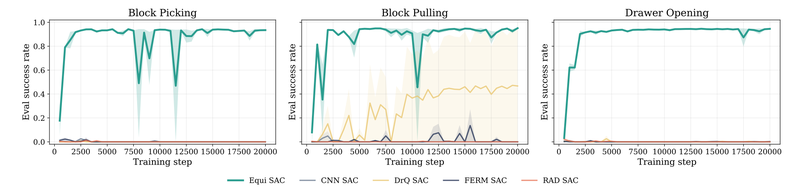
Symmetry

The planar action component $(\Delta x, \Delta y)$ transforms with the world under a rotation g; the remaining components $(\Delta z, \Delta\theta, a_{\text{grip}})$ are unchanged. The actor commutes with g, the critic is invariant to it.



The square commutes by construction: it is enforced in the weights rather than learned from rotated samples.

Results



Eval success rate vs. environment step, mean $\pm 1\sigma$ over $n{=}2$ seeds. Equi SAC (teal) is the only method that solves every task within the 20k-step budget.

	Pulling	Picking	Drawer
CNN SAC	0.000	0.000	0.001
RAD SAC	0.000	0.000	0.000
FERM SAC	0.004	0.000	0.000
DrQ SAC	0.460	0.000	0.001
Equi SAC	**0.936**	**0.923**	**0.938**

Mean eval success over the last five checkpoints (training steps 18k–20k).

Analysis

- **Sample efficiency.** Equi SAC reaches $\sim$0.93 on every task in **5k–8k** env steps; no baseline gets off the floor within 20k.
- **Architectural vs. statistical equivariance.** DrQ, RAD, and FERM try to learn rotational robustness from augmented views. At this data budget they never find the symmetry; Equi SAC has it from step 0.
- **DrQ on block pulling.** One seed climbs to $\sim$0.92, the other stays near 0. The 0.460 ± 0.460 band is seed variance between a success and a failure, not a half-learned policy.

Limitations and next steps

- **Two seeds per cell.** The 0.93-vs-0.00 gap is well beyond seed noise, but $n{=}2$ cannot rank the baselines against each other.
- **Replication, no extension yet.** We reproduce the ranking Wang et al. report (Equi SAC clears every task, CNN/augmentation baselines stall) on three tasks with two seeds each; generalization to unseen orientations is not tested.
- **Equivariant DQN is next.** The dispatcher supports `(alg=dqn,model=equi)`; a CNN-DQN baseline and the DQN trainer still need to run. Omitted here to avoid fabricated curves.
- **Group extensions.** Moving from C_8 to D_4 or continuous $SO(2)$ is a small architectural change in the e2cnn gspace.

References

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